

TP1 : Introduction to the GIT

- After installation step , we have to do the configuration step by giving the name + email address of the user . Remember it's done just one time for each user account .

```
C:\Users\Colcom2>git config --global user.name "maroua"  
C:\Users\Colcom2>git config --global user.email Maroua.AJKRIR@um6p.ma
```

- Now , let's create a directory named **tpgit (already created)** and initialize a git repository in this directory

```
C:\Users\Colcom2>mkdir tpgit  
Un sous-répertoire ou un fichier tpgit existe déjà.  
C:\Users\Colcom2>cd tpgit  
C:\Users\Colcom2\tpgit>_
```

```
C:\Users\Colcom2\tpgit>git init tpgit  
Initialized empty Git repository in C:/Users/Colcom2/tpgit/tpgit/.git/
```

- For the files created in the .git folder , we will first access this folder in the terminal , then list the elements that are inside

```

C:\Users\Colcom2\tpgit>cd .git

C:\Users\Colcom2\tpgit\.git>dir
Le volume dans le lecteur C s'appelle Windows
Le numéro de série du volume est 3C67-C5FF

Répertoire de C:\Users\Colcom2\tpgit\.git

17/09/2026  11:20    <DIR>          ..
17/09/2026  11:09                130 config
17/09/2026  11:09                73 description
17/09/2026  11:09                23 HEAD
17/09/2026  11:09    <DIR>          hooks
17/09/2026  11:09    <DIR>          info
17/09/2026  11:09    <DIR>          objects
17/09/2026  11:09    <DIR>          refs
               3 fichier(s)                226 octets
               5 Rép(s)  400.523.694.080 octets libres

```

- Now we wil create a file named fruits.txt

```

C:\Users\Colcom2\tpgit>echo. >fruits.txt

C:\Users\Colcom2\tpgit>cd .git

C:\Users\Colcom2\tpgit\.git>dir
Le volume dans le lecteur C s'appelle Windows
Le numéro de série du volume est 3C67-C5FF

Répertoire de C:\Users\Colcom2\tpgit\.git

17/09/2026  11:26    <DIR>          ..
17/09/2026  11:09                130 config
17/09/2026  11:09                73 description
17/09/2026  11:24                3 fruits.txt
17/09/2026  11:09                23 HEAD
17/09/2026  11:09    <DIR>          hooks
17/09/2026  11:09    <DIR>          info
17/09/2026  11:09    <DIR>          objects
17/09/2026  11:09    <DIR>          refs
               4 fichier(s)                229 octets
               5 Rép(s)  400.500.563.968 octets libres

```

- Now , let's make a commit , but we have to follow the usual sequence starting by the status

```
C:\Users\Colcom2\tpgit>git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    fruits.txt
    tpgit/

nothing added to commit but untracked files present (use "git add" to track)
```

- Then we will add a fruit to our file

```
C:\Users\Colcom2\tpgit>echo apple >> fruits.txt

C:\Users\Colcom2\tpgit>git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    fruits.txt
    tpgit/

nothing added to commit but untracked files present (use "git add" to track)
```

- As you can see the file was modified but its still untracked that's why we will use git add so that the changes will be in the staging area and ready to be committed

```
C:\Users\Colcom2\tpgit>git add fruits.txt

C:\Users\Colcom2\tpgit>git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   fruits.txt
```

- And now final step let's commit

```
C:\Users\Colcom2\tpgit>git commit -m "apple added"
[master (root-commit) 01ca3d6] apple added
 1 file changed, 2 insertions(+)
 create mode 100644 fruits.txt

C:\Users\Colcom2\tpgit>
```

- And we always have to make sure that everything is on track but the command status

```
C:\Users\Colcom2\tpgit>git status
On branch master
nothing to commit, working tree clean
```

- And to check the commit history with reverse chronological order we use git log

```
C:\Users\Colcom2\tpgit>git log
commit 01ca3d6ea8c90bd06cf6497eb4b462b6d273113f (HEAD -> master)
Author: maroua <Maroua.AJKRIR@um6p.ma>
Date: Thu Sep 17 11:45:35 2026 +0100

    apple added
```

- Let's do 4 additional commits, we have to make sure every commit is atomic

```
C:\Users\Colcom2\tpgit>echo orange >>fruits.txt

C:\Users\Colcom2\tpgit>git status
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   fruits.txt

no changes added to commit (use "git add" and/or "git commit -a")

C:\Users\Colcom2\tpgit>git add fruits.txt

C:\Users\Colcom2\tpgit>git status
On branch master
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   fruits.txt

C:\Users\Colcom2\tpgit>git commit -m "orange added to fruits.txt"
[master c74cb0f] orange added to fruits.txt
1 file changed, 1 insertion(+)

C:\Users\Colcom2\tpgit>git status
On branch master
nothing to commit, working tree clean

C:\Users\Colcom2\tpgit>git log
commit c74cb0f6ae14b4465b2536a38aa11af6b30225e1 (HEAD -> master)
Author: maroua <Maroua.AJKRIR@um6p.ma>
Date: Thu Sep 17 11:54:58 2026 +0100

    orange added to fruits.txt

commit 01ca3d6ea8c90bd06cf6497eb4b462b6d273113f
Author: maroua <Maroua.AJKRIR@um6p.ma>
Date: Thu Sep 17 11:45:35 2026 +0100

    apple added
```

```
C:\Users\Colcom2\tpgit>echo kiwi >>fruits.txt

C:\Users\Colcom2\tpgit>git status
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   fruits.txt

no changes added to commit (use "git add" and/or "git commit -a")

C:\Users\Colcom2\tpgit>git add fruits.txt

C:\Users\Colcom2\tpgit>git status
On branch master
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   fruits.txt

C:\Users\Colcom2\tpgit>git commit -m "kiwi added to fruits.txt"
[master 21045db] kiwi added to fruits.txt
 1 file changed, 1 insertion(+)

C:\Users\Colcom2\tpgit>git log
commit 21045db2a0578090682c258efd81c98b092b7528 (HEAD -> master)
Author: maroua <Maroua.AJKRIR@um6p.ma>
Date: Thu Sep 17 11:57:24 2026 +0100

    kiwi added to fruits.txt

commit c74cb0f6ae14b4465b2536a38aa11af6b30225e1
Author: maroua <Maroua.AJKRIR@um6p.ma>
Date: Thu Sep 17 11:54:58 2026 +0100

    orange added to fruits.txt

commit 01ca3d6ea8c90bd06cf6497eb4b462b6d273113f
Author: maroua <Maroua.AJKRIR@um6p.ma>
Date: Thu Sep 17 11:45:35 2026 +0100

    apple added
```

```
C:\Users\Colcom2\tpgit>echo grapes >>fruits.txt

C:\Users\Colcom2\tpgit>git status
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   fruits.txt

no changes added to commit (use "git add" and/or "git commit -a")

C:\Users\Colcom2\tpgit>git add fruits.txt

C:\Users\Colcom2\tpgit>git status
On branch master
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   fruits.txt

C:\Users\Colcom2\tpgit>git commit -m "grapes added to fruits.txt"
[master c52c391] grapes added to fruits.txt
 1 file changed, 1 insertion(+)

C:\Users\Colcom2\tpgit>git status
On branch master
nothing to commit, working tree clean

C:\Users\Colcom2\tpgit>git log
commit c52c3913a933ec336e6dc8fc7fee9f6265ea31c2 (HEAD -> master)
Author: maroua <Maroua.AJKRIR@um6p.ma>
Date: Thu Sep 17 11:59:14 2026 +0100

    grapes added to fruits.txt

commit 21045db2a0578090682c258efd81c98b092b7528
Author: maroua <Maroua.AJKRIR@um6p.ma>
Date: Thu Sep 17 11:57:24 2026 +0100

    kiwi added to fruits.txt

commit c74cb0f6ae14b4465b2536a38aa11af6b30225e1
Author: maroua <Maroua.AJKRIR@um6p.ma>
Date: Thu Sep 17 11:54:58 2026 +0100

    orange added to fruits.txt

commit 01ca3d6ea8c90bd06cf6497eb4b462b6d273113f
Author: maroua <Maroua.AJKRIR@um6p.ma>
Date: Thu Sep 17 11:45:35 2026 +0100

    apple added
```

```

C:\Users\Colcom2\tpgit>echo pineapple >>fruits.txt

C:\Users\Colcom2\tpgit>git status
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   fruits.txt

no changes added to commit (use "git add" and/or "git commit -a")

C:\Users\Colcom2\tpgit>git add fruits.txt

C:\Users\Colcom2\tpgit>git status
On branch master
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    modified:   fruits.txt

C:\Users\Colcom2\tpgit>git commit -m "pineapple added to fruits.txt"
[master 8dbf3b8] pineapple added to fruits.txt
 1 file changed, 1 insertion(+)

C:\Users\Colcom2\tpgit>git status
On branch master
nothing to commit, working tree clean

C:\Users\Colcom2\tpgit>git log
commit 8dbf3b86ea05560f9d76aac772adbe521a8881b8 (HEAD -> master)
Author: maroua <Maroua.AJKRIR@um6p.ma>
Date: Thu Sep 17 12:01:03 2026 +0100

    pineapple added to fruits.txt

commit c52c3913a933ec336e6dc8fc7fee9f6265ea31c2
Author: maroua <Maroua.AJKRIR@um6p.ma>
Date: Thu Sep 17 11:59:14 2026 +0100

    grapes added to fruits.txt

commit 21045db2a0578090682c258efd81c98b092b7528
Author: maroua <Maroua.AJKRIR@um6p.ma>
Date: Thu Sep 17 11:57:24 2026 +0100

    kiwi added to fruits.txt

commit c74cb0f6ae14b4465b2536a38aa11af6b30225e1
Author: maroua <Maroua.AJKRIR@um6p.ma>
Date: Thu Sep 17 11:54:58 2026 +0100

    orange added to fruits.txt

commit 01ca3d6ea8c90bd06cf6497eb4b462b6d273113f
Author: maroua <Maroua.AJKRIR@um6p.ma>
Date: Thu Sep 17 11:45:35 2026 +0100

    apple added

```

- we can also use the long command `git log --graph --oneline --decorate --all`

To have a more visual version of git log

```

C:\Users\Colcom2\tpgit>git log --graph --oneline --decorate --all
* 8dbf3b8 (HEAD -> master) pineapple added to fruits.txt
* c52c391 grapes added to fruits.txt
* 21045db kiwi added to fruits.txt
* c74cb0f orange added to fruits.txt
* 01ca3d6 apple added

```

- Once we create a branch named “vegetables we will switch to it using git checkout command

```
C:\Users\Colcom2\tpgit>git branch vergetables

C:\Users\Colcom2\tpgit>git branch
* master
  vergetables

C:\Users\Colcom2\tpgit>git checkout master
Already on 'master'

C:\Users\Colcom2\tpgit>git checkout vergetables
error: pathspec 'vergetables' did not match any file(s) known to git

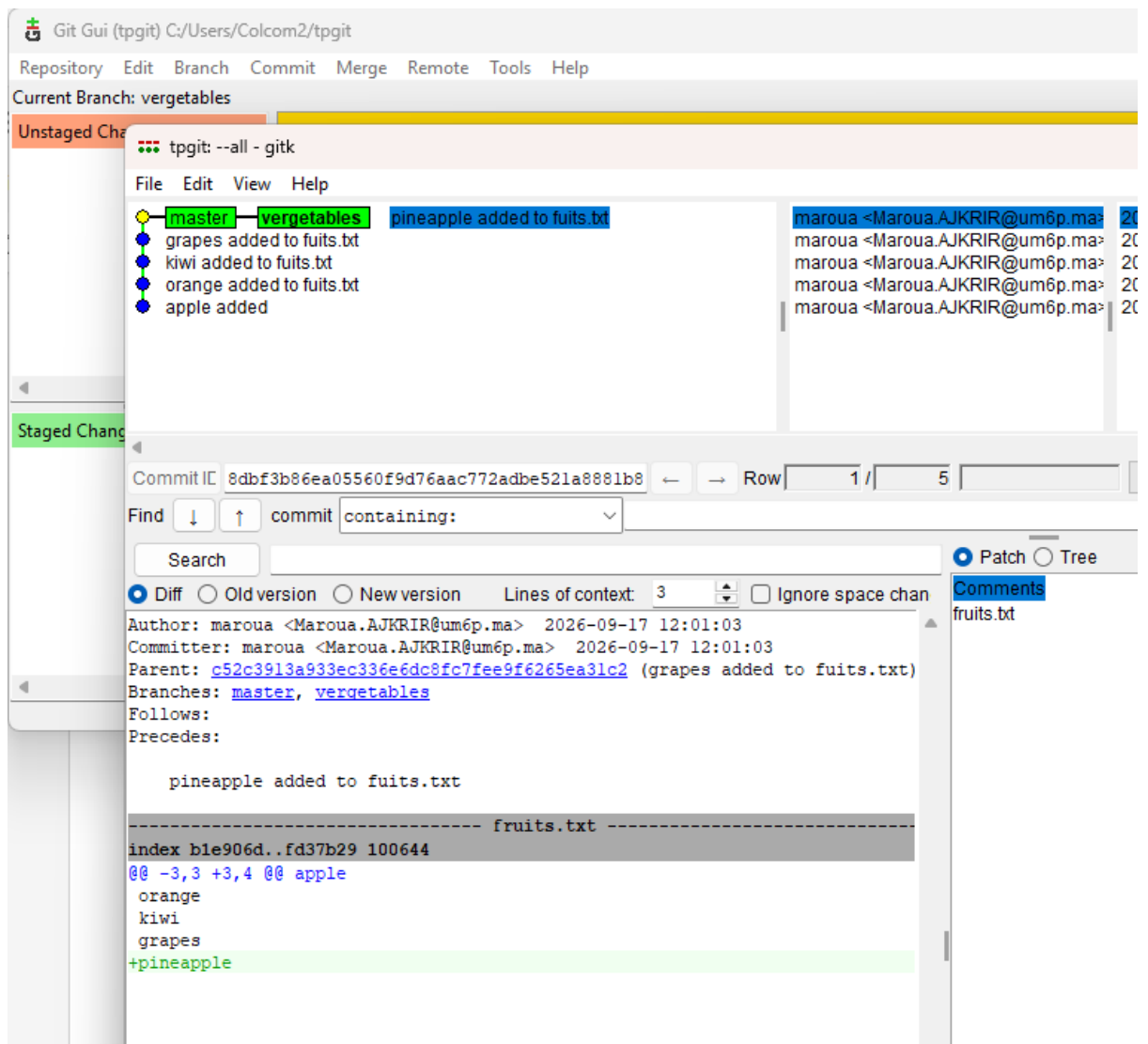
C:\Users\Colcom2\tpgit>git checkout vergetables
Switched to branch 'vergetables'

C:\Users\Colcom2\tpgit>git branch
  master
* vergetables

C:\Users\Colcom2\tpgit>
```

- And here is the graph :

```
C:\Users\Colcom2\tpgit>git gui
```



- So now let's add a file vegetables.txt in this branch

```
C:\Users\Colcom2\tpgit>echo. >vegetables.txt
C:\Users\Colcom2\tpgit>
```

- Let's add some vegetables in this file, with checking the commit history each time using git log

```
C:\Users\Colcom2\tpgit>echo potatoes >>vegetables.txt

C:\Users\Colcom2\tpgit>git status
On branch vergetables
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    vegetables.txt

nothing added to commit but untracked files present (use "git add" to track)

C:\Users\Colcom2\tpgit>git add vegetables.txt

C:\Users\Colcom2\tpgit>git status
On branch vergetables
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   vegetables.txt

C:\Users\Colcom2\tpgit>git commit -m "potaoes added to vegetables.txt"
[vergetables 0300485] potaoes added to vegetables.txt
 1 file changed, 2 insertions(+)
 create mode 100644 vegetables.txt

C:\Users\Colcom2\tpgit>git status
On branch vergetables
nothing to commit, working tree clean

C:\Users\Colcom2\tpgit>git log
commit 0300485e5939c6a7adbd5b1f7ffac850533edd6b (HEAD -> vergetables)
Author: maroua <Maroua.AJKRIR@um6p.ma>
Date: Thu Sep 17 12:30:18 2026 +0100

    potaoes added to vegetables.txt
```

```
C:\Users\Colcom2\tpgit>echo eggplant >>vegetables.txt

C:\Users\Colcom2\tpgit>git status
On branch vergetables
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   vegetables.txt

no changes added to commit (use "git add" and/or "git commit -a")

C:\Users\Colcom2\tpgit>git add vegetables.txt

C:\Users\Colcom2\tpgit>git status
On branch vergetables
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   vegetables.txt

C:\Users\Colcom2\tpgit>git commit -m "eggplant added to vegetables.txt"
[vergetables 2e659b7] eggplant added to vegetables.txt
 1 file changed, 1 insertion(+)

C:\Users\Colcom2\tpgit>git status
On branch vergetables
nothing to commit, working tree clean

C:\Users\Colcom2\tpgit>git log
commit 2e659b7175f3393bfea6d38d0954548f89a8e6b9 (HEAD -> vergetables)
Author: maroua <Maroua.AJKRIR@um6p.ma>
Date:   Thu Sep 17 12:32:15 2026 +0100

    eggplant added to vegetables.txt

commit 0300485e5939c6a7adbd5b1f7ffac850533edd6b
Author: maroua <Maroua.AJKRIR@um6p.ma>
Date:   Thu Sep 17 12:30:18 2026 +0100

    potaoes added to vegetables.txt
```

```

C:\Users\Colcom2\tpgit>echo tomato >>vegetables.txt

C:\Users\Colcom2\tpgit>git status
On branch vergetables
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   vegetables.txt

no changes added to commit (use "git add" and/or "git commit -a")

C:\Users\Colcom2\tpgit>git add vegetables.txt

C:\Users\Colcom2\tpgit>git status
On branch vergetables
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   vegetables.txt

C:\Users\Colcom2\tpgit>git commit -m "tomato added to vegetables.txt"
[vergetables 7627824] tomato added to vegetables.txt
 1 file changed, 1 insertion(+)

C:\Users\Colcom2\tpgit>git status
On branch vergetables
nothing to commit, working tree clean

C:\Users\Colcom2\tpgit>git log
commit 7627824e1e6dfc59a4941b00d60701692d8a8322 (HEAD -> vergetables)
Author: maroua <Maroua.AJKRIR@um6p.ma>
Date:   Thu Sep 17 12:33:35 2026 +0100

    tomato added to vegetables.txt

commit 2e659b7175f3393bfea6d38d0954548f89a8e6b9
Author: maroua <Maroua.AJKRIR@um6p.ma>
Date:   Thu Sep 17 12:32:15 2026 +0100

    eggplant added to vegetables.txt

commit 0300485e5939c6a7adbd5b1f7ffac850533edd6b
Author: maroua <Maroua.AJKRIR@um6p.ma>
Date:   Thu Sep 17 12:30:18 2026 +0100

    potaoes added to vegetables.txt

```

- Let's create another branch sauces

```

C:\Users\Colcom2\tpgit>git branch sauces

C:\Users\Colcom2\tpgit>git branch
  master
  sauces
* vergetables

```

```

C:\Users\Colcom2\tpgit>git checkout sauces
Switched to branch 'sauces'

```

- And add a file sauces.txt

```

C:\Users\Colcom2\tpgit>echo. >sauces.txt

```

- Let's now add two elements to sauces.txt

```

C:\Users\Colcom2\tpgit>echo. >sauces.txt

C:\Users\Colcom2\tpgit>echo tabasco >>sauces.txt

C:\Users\Colcom2\tpgit>git status
On branch sauces
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    sauces.txt

nothing added to commit but untracked files present (use "git add" to track)

C:\Users\Colcom2\tpgit>git add sauces.txt

C:\Users\Colcom2\tpgit>git status
On branch sauces
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   sauces.txt

C:\Users\Colcom2\tpgit>git commit -m "tabasco added to sauces.txt"
[sauces eafb4d6] tabasco added to sauces.txt
 1 file changed, 2 insertions(+)
 create mode 100644 sauces.txt

C:\Users\Colcom2\tpgit>git status
On branch sauces
nothing to commit, working tree clean

```

```

C:\Users\maher\tpgit>echo mayo>sauces.txt

C:\Users\maher\tpgit>git status
On branch sauces
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   sauces.txt

no changes added to commit (use "git add" and/or "git commit -a")

C:\Users\maher\tpgit>git add sauces.txt

C:\Users\maher\tpgit>git status
On branch sauces
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    modified:   sauces.txt

C:\Users\maher\tpgit>git commit -m "mayo added"
[sauces 8571bae] mayo added
 1 file changed, 1 insertion(+), 1 deletion(-)

C:\Users\maher\tpgit>git status
On branch sauces
nothing to commit, working tree clean

```

- We can check the history of commits by using git log

```

Author: maroua <Maroua.AJKRIR@um6p.ma>
Date: Thu Sep 17 21:12:04 2026 +0100

    mayo added

commit dfb3af434d25fdc60d7e9ede1c0f223433da400c
Author: maroua <Maroua.AJKRIR@um6p.ma>
Date: Thu Sep 17 21:10:33 2026 +0100

    tabasco added

commit 356bb9222634d05991f45671154e1dfe9aff569
Author: maroua <Maroua.AJKRIR@um6p.ma>
Date: Thu Sep 17 21:06:07 2026 +0100

    vegetables added

commit 793e3834f9adab72bfba31f1fc760a146ca23d9 (master)
Author: maroua <Maroua.AJKRIR@um6p.ma>
Date: Thu Sep 17 21:01:06 2026 +0100

    pineapple added

commit 9dc963f5f0269e542a834bd6968d20f9e3f2f6f5
Author: maroua <Maroua.AJKRIR@um6p.ma>
Date: Thu Sep 17 21:00:39 2026 +0100

    kiwi added

commit d027fa6d768a24d648cca146f644cb3352c81ae3
Author: maroua <Maroua.AJKRIR@um6p.ma>
Date: Thu Sep 17 20:58:59 2026 +0100

    kiwi added

commit bfa91016e45f96ebcaa4ce286104a86412ec2b2a
Author: maroua <Maroua.AJKRIR@um6p.ma>
Date: Thu Sep 17 20:58:18 2026 +0100

    orange added

commit 95cec85e3285aee34ff92832963963f6dca50691
Author: maroua <Maroua.AJKRIR@um6p.ma>
Date: Thu Sep 17 20:56:45 2026 +0100

    apple added

```

- And the branches by git branch , the actual branch has * next to it (example : sauces)

```

C:\Users\maher\tpgit>git branch
  master
* sauces
  vegetables

C:\Users\maher\tpgit>

```

- Now , we will merge the branches ,
- First let's switch to vegetables , then see if we can access sauces.txt that was created with sauces branch

```

C:\Users\maher\tpgit>git checkout vegetables
Switched to branch 'vegetables'

C:\Users\maher\tpgit>dir
Le volume dans le lecteur C s'appelle Windows
Le numéro de série du volume est 820C-9579

Répertoire de C:\Users\maher\tpgit

17/09/2026  21:20    <DIR>          .
17/09/2026  21:16    <DIR>          ..
17/09/2026  21:00                12 fruits.txt
17/09/2026  21:20                9 sauces.txt
17/09/2026  21:05                9 vegetables.txt
               3 fichier(s)                30 octets
               2 Rép(s) 277.509.967.872 octets libres

C:\Users\maher\tpgit>cd sauces.txt
Nom de répertoire non valide.

```

- So as we can see here , it's not accessible via vegetables
- Now we will merge master into vegetables ,and since no modifications have been done in master we will get “already up to date “ , but for the second we had modification in vegetables but it was successful merge

```

C:\Users\maher\tpgit>git checkout vegetables
Already on 'vegetables'

C:\Users\maher\tpgit>git merge master
Already up to date.

C:\Users\maher\tpgit>git checkout master
Switched to branch 'master'

C:\Users\maher\tpgit>git merge vegetables
Updating 793e383..7691bac
Fast-forward
 sauces.txt      | 1 +
 vegetables.txt | 1 +
 2 files changed, 2 insertions(+)
 create mode 100644 sauces.txt
 create mode 100644 vegetables.txt

C:\Users\maher\tpgit>

```

- So after merging we can access vegetables.txt via master branch , and also we will have the same commit history

```
C:\Users\maher\tpgit>git checkout master
Switched to branch 'master'
```

```
C:\Users\maher\tpgit>type vegetables.txt
eggplant
```

```
C:\Users\maher\tpgit>git log
```

```
commit 5b38992f1ac44786ac8729ec2dea726dcafb6cac (HEAD -> vegetables, master)
```

```
Author: maroua <Maroua.AJKRIR@um6p.ma>
```

```
Date: Thu Sep 17 21:43:03 2026 +0100
```

```
tomato added
```

```
commit 956ca444c51ffec80dd983264646c0bbcadd378f
```

```
Author: maroua <Maroua.AJKRIR@um6p.ma>
```

```
Date: Thu Sep 17 21:42:30 2026 +0100
```

```
potato added
```

```
commit 0b1db74912e0b2f1ac4ad75a184b62caaa170e88
```

```
Author: maroua <Maroua.AJKRIR@um6p.ma>
```

```
Date: Thu Sep 17 21:40:15 2026 +0100
```

```
eggplant added
```

```
commit 5b38992f1ac44786ac8729ec2dea726dcafb6cac (HEAD -> master, vegetables)
```

```
Author: maroua <Maroua.AJKRIR@um6p.ma>
```

```
Date: Thu Sep 17 21:43:03 2026 +0100
```

```
tomato added
```

```
commit 956ca444c51ffec80dd983264646c0bbcadd378f
```

```
Author: maroua <Maroua.AJKRIR@um6p.ma>
```

```
Date: Thu Sep 17 21:42:30 2026 +0100
```

```
potato added
```

```
commit 0b1db74912e0b2f1ac4ad75a184b62caaa170e88
```

```
Author: maroua <Maroua.AJKRIR@um6p.ma>
```

```
Date: Thu Sep 17 21:40:15 2026 +0100
```

```
eggplant added
```

- Let's do the same for sauces

```
C:\Users\maher\tpgit>git checkout sauces
Switched to branch 'sauces'
```

```
C:\Users\maher\tpgit>git merge master
Already up to date.
```

```
C:\Users\maher\tpgit>git checkout master
Switched to branch 'master'
```

```
C:\Users\maher\tpgit>git merge sauces
Updating 5b38992..c0891f0
Fast-forward
 sauces.txt | 2 ++
 1 file changed, 2 insertions(+)
```

```
C:\Users\maher\tpgit>git checkout master
Switched to branch 'master'
```

```
C:\Users\maher\tpgit>type sauces.txt

tabasco
mayo
```

```
C:\Users\maher\tpgit>git log
commit c0891f0659de15b9a2844c6d4ae7e5a76de5e459 (HEAD -> master, sauces)
Merge: 7e30c1e 5b38992
Author: maroua <Maroua.AJKRIR@um6p.ma>
Date: Thu Sep 17 21:59:07 2026 +0100
```

-

```
commit 7e30c1eb226f5a674a31f7e51b1d0fa55e773a5c
Author: maroua <Maroua.AJKRIR@um6p.ma>
Date: Thu Sep 17 21:51:48 2026 +0100
```

tabasco and mayo added

```
commit c0891f0659de15b9a2844c6d4ae7e5a76de5e459 (HEAD -> sauces, master)
Merge: 7e30c1e 5b38992
Author: maroua <Maroua.AJKRIR@um6p.ma>
Date: Thu Sep 17 21:59:07 2026 +0100
```

-

```
commit 7e30c1eb226f5a674a31f7e51b1d0fa55e773a5c
Author: maroua <Maroua.AJKRIR@um6p.ma>
Date: Thu Sep 17 21:51:48 2026 +0100
```

tabasco and mayo added

- All the branches are up to date : as we can see everything is merged to master


```

Colcom9@FDELL-5G9Q4H4 MINGW64 ~
$ cd tp_maroua_yazid/

Colcom9@FDELL-5G9Q4H4 MINGW64 ~/tp_maroua_yazid (master)
$ git log
commit be720e30d9bb4569969e0245f834aab0109cd337 (HEAD -> master, origin/master, origin/HEAD)
Author: maroua <Maroua.AJKRIR@um6p.ma>
Date: Thu Sep 17 22:37:29 2026 +0100

    mayo added

commit 8cb18a75b913ed7524f642342e26de997cb33e95
Author: maroua <Maroua.AJKRIR@um6p.ma>
Date: Thu Sep 17 22:37:04 2026 +0100

    tabasco added

commit 8d997fca67f2990585f08dcf880d42fde33e4fe7
Author: maroua <Maroua.AJKRIR@um6p.ma>
Date: Thu Sep 17 22:35:19 2026 +0100

    tomato added

commit 5335d67a7b19712ce683e3114fb09f48d4ab5b11
Author: maroua <Maroua.AJKRIR@um6p.ma>
Date: Thu Sep 17 22:34:55 2026 +0100

    eggplant added

commit 8b30fff79370b3224c304ec79c6687dd60bec8c1
Author: maroua <Maroua.AJKRIR@um6p.ma>
Date: Thu Sep 17 22:34:23 2026 +0100

    potatoes added

commit 02956bd2bdb21855eccc81ef23eb7bd206824b70
Author: maroua <Maroua.AJKRIR@um6p.ma>
Date: Thu Sep 17 22:32:02 2026 +0100

    pineapple added

commit 173b862ec6996bb142a615e056765b9075e6763e
Author: maroua <Maroua.AJKRIR@um6p.ma>
Date: Thu Sep 17 22:30:51 2026 +0100

    grapes added

commit 392d63184ad7e08ff48d71358564aa0923f49f92
Author: maroua <Maroua.AJKRIR@um6p.ma>
Date: Thu Sep 17 22:30:26 2026 +0100

    kiwi added

commit d7ad715b4bb2e193e391871e90760a40116f61d2
Author: maroua <Maroua.AJKRIR@um6p.ma>
Date: Thu Sep 17 22:29:55 2026 +0100

    orange added

commit 1a61e5220b70cdb3a2319ae55e3790aa532692a4
Author: maroua <Maroua.AJKRIR@um6p.ma>
Date: Thu Sep 17 22:29:20 2026 +0100

    apple added

```

- Merge without conflicts , work of student 1

```
C:\Users\maher\tpgit1>git branch spices

C:\Users\maher\tpgit1>git branch
* master
  sauces
  spices
  vegetables

C:\Users\maher\tpgit1>git checkout spices
Switched to branch 'spices'

C:\Users\maher\tpgit1>echo .>>spices.txt

C:\Users\maher\tpgit1>git add spices.txt

C:\Users\maher\tpgit1>git commit -m "create spices.txt"
[spices e616c3a] create spices.txt
1 file changed, 1 insertion(+)
create mode 100644 spices.txt
```

```
C:\Users\maher\tpgit1>echo cumin>>spices.txt

C:\Users\maher\tpgit1>git add spices.txt

C:\Users\maher\tpgit1>git commit -m "add cumin"
[spices 9da6443] add cumin
1 file changed, 1 insertion(+)
```

```
C:\Users\maher\tpgit1>echo salt>>spices.txt

C:\Users\maher\tpgit1>git add spices.txt

C:\Users\maher\tpgit1>git commit -m "add salt"
[spices b8f8111] add salt
1 file changed, 1 insertion(+)
```

```

C:\Users\maher\tpgit1>git merge master
Already up to date.

C:\Users\maher\tpgit1>git checkout master
Switched to branch 'master'
Your branch is up to date with 'origin/master'.

C:\Users\maher\tpgit1>git merge spices
Updating be720e3..b8f8111
Fast-forward
 spices.txt | 3 +++
 1 file changed, 3 insertions(+)
 create mode 100644 spices.txt

```

```

C:\Users\maher\tpgit1>git pull
remote: Enumerating objects: 7, done.
remote: Counting objects: 100% (7/7), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 6 (delta 1), reused 6 (delta 1), pack-reused 0 (from 0)
Unpacking objects: 100% (6/6), 532 bytes | 19.00 KiB/s, done.
From https://github.com/marvouch/tpgit
 be720e3..1662097 master -> origin/master
Merge made by the 'ort' strategy.
 herbs.txt | 2 ++
 1 file changed, 2 insertions(+)
 create mode 100644 herbs.txt

C:\Users\maher\tpgit1>git push
Enumerating objects: 13, done.
Counting objects: 100% (13/13), done.
Delta compression using up to 24 threads
Compressing objects: 100% (8/8), done.
Writing objects: 100% (11/11), 943 bytes | 471.00 KiB/s, done.
Total 11 (delta 4), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (4/4), completed with 1 local object.
To https://github.com/marvouch/tpgit
 1662097..d8d8140 master -> master

```

```

C:\Users\maher\tpgit1>ls
'ls' n'est pas reconnu en tant que commande interne
ou externe, un programme exécutable ou un fichier de commandes.

```

```

C:\Users\maher\tpgit1>dir
Le volume dans le lecteur C s'appelle Windows
Le numéro de série du volume est 820C-9579

```

Répertoire de C:\Users\maher\tpgit1

```

18/09/2026  14:23    <DIR>          .
18/09/2026  14:25    <DIR>          ..
17/09/2026  22:31                42 fruits.txt
18/09/2026  14:23                18 herbs.txt
17/09/2026  22:40                17 sauces.txt
18/09/2026  14:18                16 spices.txt
17/09/2026  22:39                30 vegetables.txt
                5 fichier(s)                123 octets
                2 Rép(s)  273.215.340.544 octets libres

```

```

C:\Users\maher\tpgit1>

```

- Work of student 2:

```

Colcom9@FDELL-5G9Q4H4 MINGW64 ~/tp_maroua_yazid (master)
$ git branch herbs

Colcom9@FDELL-5G9Q4H4 MINGW64 ~/tp_maroua_yazid (master)
$ cd herbs
bash: cd: herbs: No such file or directory

Colcom9@FDELL-5G9Q4H4 MINGW64 ~/tp_maroua_yazid (master)
$ git checkout herbs
Switched to branch 'herbs'

Colcom9@FDELL-5G9Q4H4 MINGW64 ~/tp_maroua_yazid (herbs)
$ touch herbs.txt

Colcom9@FDELL-5G9Q4H4 MINGW64 ~/tp_maroua_yazid (herbs)
$ /echo mint>>herbs.txt
bash: /echo: No such file or directory

Colcom9@FDELL-5G9Q4H4 MINGW64 ~/tp_maroua_yazid (herbs)
$ echo mint>>herbs.txt

Colcom9@FDELL-5G9Q4H4 MINGW64 ~/tp_maroua_yazid (herbs)
$ git add herbs.txt
warning: in the working copy of 'herbs.txt', LF will be replaced by CRLF the next time Git touches it

Colcom9@FDELL-5G9Q4H4 MINGW64 ~/tp_maroua_yazid (herbs)
$ git commit -m "kfc"
[herbs 301e342] kfc
1 file changed, 1 insertion(+)
create mode 100644 herbs.txt

Colcom9@FDELL-5G9Q4H4 MINGW64 ~/tp_maroua_yazid (herbs)
$ echo corriander>>herbs.txt

Colcom9@FDELL-5G9Q4H4 MINGW64 ~/tp_maroua_yazid (herbs)
$ cat herbs.txt
mint
corriander

Colcom9@FDELL-5G9Q4H4 MINGW64 ~/tp_maroua_yazid (herbs)
$ git add herbs.txt
warning: in the working copy of 'herbs.txt', LF will be replaced by CRLF the next time Git touches it

Colcom9@FDELL-5G9Q4H4 MINGW64 ~/tp_maroua_yazid (herbs)
$ git commit -m "herbs2"
[herbs 1662097] herbs2
1 file changed, 1 insertion(+)

```

-we checked if we had the same elements (commits)

```
C:\Users\maher\tpgit1>git log
commit d8d8140c6601eaa603dbd592738d25733f4e55dd (HEAD -> master, origin/master, origin/HEAD)
Merge: b8f8111 1662097
Author: maroua <Maroua.AJKRIR@um6p.ma>
Date:   Fri Sep 18 14:23:00 2026 +0100

    Merge branch 'master' of https://github.com/marvouch/tpgit

commit b8f81113e0f2acc2864083519a0b85030ee59421 (spices)
Author: maroua <Maroua.AJKRIR@um6p.ma>
Date:   Fri Sep 18 14:14:42 2026 +0100

    add salt

commit 16620978b70c814d2ed3c2e3aea4f55b870a188c
Author: Yazid <yazid.arezdi@um6p.com>
Date:   Fri Sep 18 14:13:57 2026 +0100

    herbs2

commit 301e342473290769e15c064860f33df1956c77cc
Author: Yazid <yazid.arezdi@um6p.com>
Date:   Fri Sep 18 14:12:51 2026 +0100

    kfc

commit 9da6443d68844a005f8095ec077d270ab43365a0
Author: maroua <Maroua.AJKRIR@um6p.ma>
Date:   Fri Sep 18 14:12:14 2026 +0100

    add cumin

commit e616c3a5e5535c6c8daf045fcc1fd2c9f37e6de4
Author: maroua <Maroua.AJKRIR@um6p.ma>
Date:   Fri Sep 18 14:08:26 2026 +0100

    create spices.txt
```

```
commit d8d8140c6601eaa603dbd592738d25733f4e55dd (HEAD -> master, origin/master, origin/HEAD)
Merge: b8f8111 1662097
Author: maroua <Maroua.AJKRIR@um6p.ma>
Date: Fri Sep 18 14:23:00 2026 +0100

    Merge branch 'master' of https://github.com/marvouch/tpgit

commit b8f81113e0f2acc2864083519a0b85030ee59421
Author: maroua <Maroua.AJKRIR@um6p.ma>
Date: Fri Sep 18 14:14:42 2026 +0100

    add salt

commit 16620978b70c814d2ed3c2e3aea4f55b870a188c (herbs)
Author: Yazid <yazid.arezdi@um6p.com>
Date: Fri Sep 18 14:13:57 2026 +0100

    herbs2

commit 301e342473290769e15c064860f33df1956c77cc
Author: Yazid <yazid.arezdi@um6p.com>
Date: Fri Sep 18 14:12:51 2026 +0100

    kfc

commit 9da6443d68844a005f8095ec077d270ab43365a0
Author: maroua <Maroua.AJKRIR@um6p.ma>
Date: Fri Sep 18 14:12:14 2026 +0100

    add cumin

commit e616c3a5e5535c6c8daf045fcc1fd2c9f37e6de4
Author: maroua <Maroua.AJKRIR@um6p.ma>
Date: Fri Sep 18 14:08:26 2026 +0100

    create spices.txt

commit be720e30d9bb4569969e0245f834aab0109cd337
Author: maroua <Maroua.AJKRIR@um6p.ma>
Date: Thu Sep 17 22:37:29 2026 +0100

    mayo added

commit 8cb18a75b913ed7524f642342e26de997cb33e95
Author: maroua <Maroua.AJKRIR@um6p.ma>
Date: Thu Sep 17 22:37:04 2026 +0100

    tabasco added

commit 8d997fca67f2990585f08dcf880d42fde33e4fe7
Author: maroua <Maroua.AJKRIR@um6p.ma>
Date: Thu Sep 17 22:35:19 2026 +0100

    tomato added

commit 5335d67a7b19712ce683e3114fb09f48d4ab5b11
Author: maroua <Maroua.AJKRIR@um6p.ma>
Date: Thu Sep 17 22:34:55 2026 +0100

    eggplant added

commit 8b30fff79370b3224c304ec79c6687dd60bec8c1
Author: maroua <Maroua.AJKRIR@um6p.ma>
Date: Thu Sep 17 22:34:23 2026 +0100

    potatoes added

commit 02956bd2bdb21855eccc81cf23eb7bd206824b70
Author: maroua <Maroua.AJKRIR@um6p.ma>
Date: Thu Sep 17 22:32:02 2026 +0100

    ;|
```

←

↺

https://github.com/marvouch/tpgit

🔍

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🍴 master

🍴

📄

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<> Code

 marvouch

Merge branch 'master' of <https://github.com/marvouch/tpgit>

d8d8140 · 11 minutes ago

🕒

📄 fruits.txt	pineapple added	16 hours ago
📄 herbs.txt	herbs2	20 minutes ago
📄 sauces.txt	mayo added	16 hours ago
📄 spices.txt	add salt	19 minutes ago
📄 vegetables.txt	tomato added	16 hours ago

 marvouch

Merge branch 'master' of <https://github.com/marvouch/tpgit>

d8d8140 · 11 minutes ago

🕒 16 Commits

📄 fruits.txt	pineapple added	16 hours ago
📄 herbs.txt	herbs2	20 minutes ago
📄 sauces.txt	mayo added	16 hours ago
📄 spices.txt	add salt	19 minutes ago
📄 vegetables.txt	tomato added	16 hours ago

- Merge with conflict :
(work done by maroua) :

```
C:\Users\maher\tpgit1>git push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 24 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 309 bytes | 309.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/marvouch/tpgit
    d8d8140..855f0a5  master -> master

C:\Users\maher\tpgit1>
```

```
C:\Users\maher\tpgit1>git push
To https://github.com/marvouch/tpgit
 ! [rejected]        master -> master (fetch first)
error: failed to push some refs to 'https://github.com/marvouch/tpgit'
hint: Updates were rejected because the remote contains work that you do not
hint: have locally. This is usually caused by another repository pushing to
hint: the same ref. If you want to integrate the remote changes, use
hint: 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.

C:\Users\maher\tpgit1>git merge --abort
fatal: There is no merge to abort (MERGE_HEAD missing).

C:\Users\maher\tpgit1>git pull
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (1/1), done.
remote: Total 3 (delta 1), reused 3 (delta 1), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 251 bytes | 27.00 KiB/s, done.
From https://github.com/marvouch/tpgit
   b601f25..504acec  master       -> origin/master
Updating b601f25..504acec
Fast-forward
 vegetables.txt | 1 -
 1 file changed, 1 deletion(-)

C:\Users\maher\tpgit1>
```

Work done by Yazid :

```

Colcom9@FDELL-5G9Q4H4 MINGW64 ~/tp_maroua_yazid (master)
$ cat fruits.txt

apple
orange
kiwi
grapes
pineapple
cherries

Colcom9@FDELL-5G9Q4H4 MINGW64 ~/tp_maroua_yazid (master)
$ git add fruits.txt
warning: in the working copy of 'fruits.txt', LF will be replaced by CRLF the next time Git touches it

Colcom9@FDELL-5G9Q4H4 MINGW64 ~/tp_maroua_yazid (master)
$ git commit -m "aiaiaiaaaa"
[master 32f2f7b] aiaiaiaaaa
1 file changed, 1 insertion(+)

Colcom9@FDELL-5G9Q4H4 MINGW64 ~/tp_maroua_yazid (master)
$ git pull
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (1/1), done.
remote: Total 3 (delta 1), reused 3 (delta 1), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 289 bytes | 20.00 KiB/s, done.
From https://github.com/marvouch/tpgit
   d8d8140..855f0a5 master    -> origin/master
Auto-merging fruits.txt
CONFLICT (content): Merge conflict in fruits.txt
Automatic merge failed; fix conflicts and then commit the result.

Colcom9@FDELL-5G9Q4H4 MINGW64 ~/tp_maroua_yazid (master)
$ git pull
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (1/1), done.
remote: Total 3 (delta 1), reused 3 (delta 1), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 289 bytes | 20.00 KiB/s, done.
From https://github.com/marvouch/tpgit
   d8d8140..855f0a5 master    -> origin/master
Auto-merging fruits.txt
CONFLICT (content): Merge conflict in fruits.txt
Automatic merge failed; fix conflicts and then commit the result.

Colcom9@FDELL-5G9Q4H4 MINGW64 ~/tp_maroua_yazid (master|MERGING)
$ git fetch

Colcom9@FDELL-5G9Q4H4 MINGW64 ~/tp_maroua_yazid (master|MERGING)
$ git fetch origin

Colcom9@FDELL-5G9Q4H4 MINGW64 ~/tp_maroua_yazid (master|MERGING)
$ ls
fruits.txt  herbs.txt  sauces.txt  spices.txt  vegetables.txt

Colcom9@FDELL-5G9Q4H4 MINGW64 ~/tp_maroua_yazid (master|MERGING)
$ cat fruits.txt

apple
orange
kiwi
grapes
pineapple
<<<<<<< HEAD
cherries
=====
banana
>>>>>>> 855f0a54c7c60e047bddde017dae087d616cbe7e

Colcom9@FDELL-5G9Q4H4 MINGW64 ~/tp_maroua_yazid (master|MERGING)
$

```

```
Colcom9@FDELL-5G9Q4H4 MINGW64 ~/tp_maroua_yazid (master)
$ ls
fruits.txt  herbs.txt  sauces.txt  spices.txt  vegetables.txt

Colcom9@FDELL-5G9Q4H4 MINGW64 ~/tp_maroua_yazid (master)
$ git pull
Already up to date.

Colcom9@FDELL-5G9Q4H4 MINGW64 ~/tp_maroua_yazid (master)
$ ls
fruits.txt  herbs.txt  sauces.txt  spices.txt  vegetables.txt

Colcom9@FDELL-5G9Q4H4 MINGW64 ~/tp_maroua_yazid (master)
$ git push
Everything up-to-date

Colcom9@FDELL-5G9Q4H4 MINGW64 ~/tp_maroua_yazid (master)
$ git add fruits.txt
warning: in the working copy of 'fruits.txt', LF will be replaced by CRLF the next time Git touches it

Colcom9@FDELL-5G9Q4H4 MINGW64 ~/tp_maroua_yazid (master)
$ git commit -m "mybad"
[master 773fbeb] mybad
1 file changed, 6 insertions(+)
create mode 100644 fruits.txt

Colcom9@FDELL-5G9Q4H4 MINGW64 ~/tp_maroua_yazid (master)
$ git push
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 14 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 297 bytes | 297.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/marvouch/tpgit.git
99b1b9b..773fbeb master -> master
```

```
Colcom9@FDELL-5G9Q4H4 MINGW64 ~/tp_maroua_yazid (master)
$ git commit -m "iamgoat"
[master c0916cd] iamgoat
1 file changed, 1 insertion(+), 1 deletion(-)

Colcom9@FDELL-5G9Q4H4 MINGW64 ~/tp_maroua_yazid (master)
$ git push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 14 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 271 bytes | 271.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/marvouch/tpgit.git
773fbeb..c0916cd master -> master

Colcom9@FDELL-5G9Q4H4 MINGW64 ~/tp_maroua_yazid (master)
$ cat vegetables.txt

potatoes
carrot
tomato
```

```

Colcom9@FDELL-5G9Q4H4 MINGW64 ~/tp_maroua_yazid (master)
$ git push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 14 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 277 bytes | 277.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/marvouch/tpgit.git
   387825d..b601f25  master -> master

Colcom9@FDELL-5G9Q4H4 MINGW64 ~/tp_maroua_yazid (master)
$ nano vegetables.txt

Colcom9@FDELL-5G9Q4H4 MINGW64 ~/tp_maroua_yazid (master)
$ git add vegetables.txt

Colcom9@FDELL-5G9Q4H4 MINGW64 ~/tp_maroua_yazid (master)
$ git commit -m "chehm"
[master 504acec] chehm
1 file changed, 1 deletion(-)

Colcom9@FDELL-5G9Q4H4 MINGW64 ~/tp_maroua_yazid (master)
$ git push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 14 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 271 bytes | 271.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/marvouch/tpgit.git
   b601f25..504acec  master -> master

```

- And here is my readme

tpgit / README.md

 **marvouch** Add READ.me

Preview

Code

Blame

11 lines (8 loc) · 198 Bytes

TP Introduction to GIT

Description

This is a list of ingredients, our first attempt at working with git

Authors

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Objectives

- getting used to git commands

THANK YOU,